

However, some researchers have shown a simple process for coating standard cotton threads with carbon nanotubes. Being much thicker than nanotube yarns, such threads can be woven more easily. The researchers dispersed carbon nanotubes in a dilute solution of a mixture of Nafion—a commercial synthetic polymer—and ethanol. They then repeatedly dipped cotton threads, 1.5mm in diameter, into the solution, letting them dry between each dip. This allowed the nanotubes to cover individual cotton strands and to adhere strongly to the surface of the cellulose fibres in the strands. The

Cotton threads can be transformed into intelligent e-textiles by coating them with carbon nanotubes

process also encouraged nanotubes to arrange themselves along the axis of cotton fibres, which increased the electrical connectivity. After several dips, it was found that cotton threads were conductive enough to wire up a light-emitting diode.

In a further test, the researchers added molecules of a material that reacts with human serum albumin to the dipping solution. Then they immersed more cotton threads. This time they ran an electrical current through the thread while exposing it to different concentrations of albumin. They found that the threads' electrical conductivity varied according to the level of albumin. The researchers propose that such material could be used to detect bleeding if suitably woven into clothing.

Expanding applications

Intelligent clothing that adapts to changing conditions could provide adjustable levels of protection from microbes, chemicals, radiation, etc. Commercial manufacturers see huge potential in clothes that glow, do not wrinkle or overcome body odour.

Materials can already be made to do some of these things, but they are too bulky, rigid or complicated for practical use. So the aim is to manufacture a light material that can be easily woven, is highly durable and, in order to transmit information, can conduct electricity.

Described below are a few proof-of-concept demonstrations that provide a direct pathway for application of these materials as wearable bi-monitoring and telemedicine sensors that are simple, sensitive, selective and versatile.

Body temperature control. Smart clothing that adapts to changing temperatures in order to keep the wearer comfortable is being developed using nature as a guide. The latest in micro technology will be used to produce material that will let air in to cool the wearer when it is hot and shut air out when it is cold. This is similar to a system used by pine cones to open up and emit seeds.

These smart garments will have a top layer of tiny spikes of water-absorbent material, possibly wool, each only 1/200th of a millimetre wide. When the wearer of the clothing sweats, the tiny spikes in the material react to the moisture and automatically open up so that air from the outside can get through the material to cool the wearer. When the wearer stops sweating, the spikes close down again to stop air from getting in. The lower layer will be made of a non-porous material so that rain can never get through from the outside, whether the spikes are open or closed.

Fabrics to monitor health. Researchers are working on an underwear, bed clothes and a T-shirt that can gather information on the health condition of the wearer.

The underwear will have four sensors placed around the waist that will collect information to calculate the speed of blood flow in arteries. Healthy blood vessels have elastic properties and the blood runs relatively slow while it presses the arterial walls.

The bed clothes include sheets, quilts and pillows. These will record the levels of blood chemistry, heart rhythm and perspiration while the user is sleeping. The pillow and the sheet will measure his heart frequency (neck, ears and feet). Also, there is a device that will measure the breathing rate in the middle of the bed.

As mentioned earlier, researchers have been able to produce carbon nanotube-coated cotton fibres capable of detecting blood and signaling its presence electrically. Clothing made of these fibres could be useful in high-risk professions to detect bleeding.

Researchers are also investigating use of the same principle to create a belt that monitors a baby's heart rate. They are investigating the ability of optical fibres to act as a sensor. If optical fibres were incorporated into clothes, they could measure the oxygen content of blood and monitor patients in MRI chambers. The project involves a series of sensors designed to analyse body sweat. Ionic biosensors measure the sodium, potassium and chloride levels; pH sensors measure the acidity; circuits measure the conductivity; and an immunosensor measures the proteins. Together, these sensors can perform diagnostics on the wearer, as well as collect data about the energy that the user exerts throughout the day.

Firefighter safety clothing with thermal sensor. Intelligent clothing has added value in professional clothing as well. The most innovative use of smart textiles is a firefighter jacket with thermal sensor technology integrated directly into the fabric. This innovative Turnout Gear is used by professional fire-fighters across the globe.

The difference between pain and second-degree burn is a mere 12°C (54°F) increase in skin temperature. Integrating thermal sensor technology directly into the fabric layers enables visual indication of critical heat levels to the firefighter in action and his colleagues before it's too late.